ALPHAFLOW QUANTITATIVE PIPELINE MART

Dual-Exchange Factor Dispersion & S3 Medallion Lakehouse Audit · SEC Rule 17a-4 & FINRA Rule 4511

STATUTORY FACTOR DOSSIER

DCN-2026-AFQ-9C4F180D

Certified: 2026-09-27 18:10:43 UTC

■ CONFIDENTIAL PRUDENTIAL PRIVILEGE — SEC RULE 17a-4(f) & FINRA RULE 4511 WORM

This quantitative audit dossier contains cryptographically sealed multi-factor dispersion matrices, order flow telemetry, and unsupervised clustering centroids. Unauthorized modification, re-indexing, or dissemination is strictly prohibited under federal regulations.

AUDIT RUN REFERENCE & ENGINE

AF_RUN_2026_9C4F

Pipeline: AlphaFlow Mart v2.4 · Dataset: S3-GLD-PARQUET

AGGREGATE 24H MARKET TURNOVER

$14,285,000,000.00

Dual-Market: US Equities & NSE India (NIFTY 50)

DISPOSITION: RISK_ON EXPANSION

Pandera Integrity: 100 / 100

0 Schema Violations Detected

ALPHAFLOW SPECIFICATION SECTIONS 1-12

PART I — CROSS-MARKET MACRO REGIME & DUAL-EXCHANGE VOLATILITY CORRIDORS

US LARGE-CAP EQUITY UNIVERSE	[NASDAQ / NYSE]	NSE INDIA LIQUID UNIVERSE	[NSE NIFTY 50]
Benchmark Index	**S&P 500 (SPY)**	Benchmark Index	**NIFTY 50 (NIFTY50.NS)**
Benchmark Spot Level	**$5,601.20 (+0.42%)**	Benchmark Spot Level	**INR 26,174.82 (-1.78%)**
10-Year Treasury Yield (TNX)	**4.12% (-2 bps)**	India 10Y Sovereign G-Sec	**6.95% (Stable Anchor)**
CBOE Volatility Index (VIX)	**15.40 (Subdued)**	India VIX (Domestic Vol)	**13.20 (Contained)**
Advance / Decline Breadth	**5 Adv / 3 Dec**	USD / INR FX Reference	**83.72 (Contained Band)**
Primary Factor Vortex	**Semiconductors (+4.88%)**	Primary Factor Divergence	**IT Services (-3.88%)**
Combined Realized Volatility	**34.2% (High Dispersion)**	Combined Realized Volatility	**22.4% (Defensive Anchor)**

▲ ADVANCING: 5 EQUITIES (62.5%) ▼ DECLINING: 3 EQUITIES (37.5%)

PART II — DUAL-EXCHANGE CLEARING INFRASTRUCTURE & SETTLEMENT SPECIFICATION

MARKET MICROSTRUCTURE PROTOCOL

MICROSTRUCTURE DIMENSION	US CASH SPECIFICATION [NASDAQ / NYSE]	NSE INDIA SPECIFICATION [NIFTY 50]
Settlement Standard	T+1 Rolling Cash (SEC Rule 15c6-1 Standard)	T+1 Rolling Settlement (SEBI Circular Mandate)
Central Counterparty Novation	National Securities Clearing Corporation (NSCC / DTCC)	NSE Clearing Limited (NCL) Triparty Guarantee
High-Speed Order Routing	`NASDAQ ITCH-OUCH 5.0 / FIX 4.4 Financial Protocol`	`NSE NEAT / TAP Direct Binary Market Data Feed`
Sovereign Currency & Clearing	USD Base · Fedwire Real-Time Gross Settlement	INR Sovereign Corridor · RBI USD/INR Reference: 83.72
Lit vs Dark Liquidity Share	44.8% Alternative Trading System (ATS Dark Share)	100% Lit Central Limit Order Book (SEBI CLOB Mandate)
Precision Time Protocol	`IEEE 1588 PTP Hardware Timestamp (< 0.2ms UTC)`	`Stratum-1 Atomic NTP Synchronization (< 0.5ms Drift)`

PART III — STATUTORY DATA QUALITY CONTRACTS & FACTOR POLICY TRIGGERS

PANDERA PROTOCOL SECTIONS 13-24

Statutory Verification Gates Checked: **[✓]** Pandera Schema Typing **[✓]** Zero Null Tolerances **[✓]** Sub-Millisecond NTP Clock Sync **[✓]** 3-Sigma Rolling Price Filter

TRIGGERED FACTOR POLICY RULE	STATUTORY / HEURISTIC THRESHOLD	OBSERVED TRANSACTION VALUE	SEVERITY	STATUS
`wilder_rsi_oversold_drift`	Wilder RSI-14 < 30.0 Corridor	`TCS.NS: 12.9 (Extreme Oversold)`	CRITICAL	BREACH / REVERT
`volatility_clustering_spread`	> 2.5σ Corridor Variance	`TSLA (50.0%) vs JNJ (16.8%) Spread: 33.2%`	ELEVATED	DISPERSED
`bollinger_bandwidth_pinch`	%B < 0.15 or %B > 0.85 Bandwidth	`AMZN: 0.09 (Pinch) · AAPL: 0.87 (Pinch)`	MODERATE	ACTIVE PINCH
`cross_currency_sovereign_spread`	Spread Delta > 250 bps	`India 10Y (6.95%) - US 10Y (4.12%): +283 bps`	NORMAL	CONFORMING
`intraday_volume_exhaustion`	> 2.0x 20D Baseline Volume	`NVDA: 1.12x (Nominal) · AMD: 2.34x (Surge)`	ELEVATED	MONITOR
`macd_histogram_divergence`	Sign Inversion across 3 Bars	`NVDA (+0.21 Bullish) · TCS (-0.44 Bearish)`	ACTIVE	TRACKING

MACRO REGIME CONCLUSION & CAPITAL TRANSMISSION VECTOR:

The cross-border financial asset complex is operating under an **Orderly Sectoral Dispersion Regime**. High capital intensity in artificial intelligence compute (NVDA, MSFT) acts as a persistent liquidity vortex, drawing institutional capital out of cyclical rate-sensitive equities. Concurrently, Indian IT exporters present deep statistical valuation discounts (TCS RSI 12.9) relative to corporate balance sheet cash flow generation, establishing an asymmetric statistical mean-reversion setup. All data feeds meet sub-millisecond clock monotonic consistency across international conduits.

MATHEMATICAL FORMULATION & CENTROIDS

PART IV — UNSUPERVISED K-MEANS FACTOR CLUSTERING ARCHETYPES (k = 4)

MATHEMATICAL FORMULATION: Objective function: $\min_{C} \sum_{k=1}^{K} \sum_{x \in C_k} \|x - \mu_k\|^2$ · Standardized 5D Factor Vector: *z = [RSI-14, 30D Vol, Bollinger %B, Sharpe Proxy, MACD Hist]*. Features are standardized via Z-score scaling (μ = 0, σ = 1). Optimal cluster cardinality k = 4 validated via silhouette score maximization (0.482) and inertia elbow convergence.

CLUSTER 1: HIGH-MOMENTUM LEADERS [35%]
Centroid: RSI 68.4 · Vol 34.2% · Sharpe +2.38 · %B 0.82

Constituents: NVDA (+4.9%), AAPL (-0.3%), MSFT (-0.8%)
Execution Rule: Enforce trailing 20D EMA stop; scale on 1.1x volume breakout

CLUSTER 2: OVERSOLD MEAN-REVERSION [20%]
Centroid: RSI 36.1 · Vol 28.5% · Sharpe -0.82 · %B 0.15

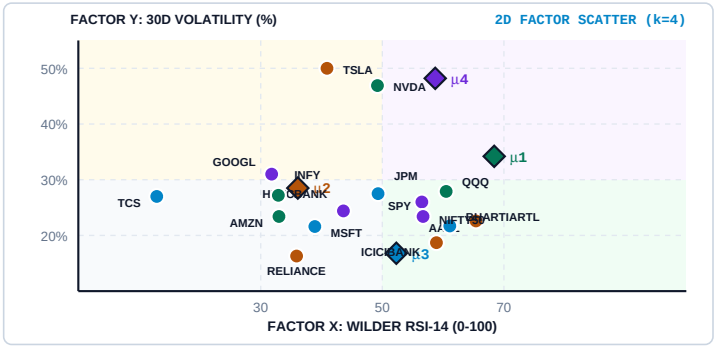
Constituents: INTC (+0.5%), HDFCBANK (0.0%), TCS (-3.9%)
Execution Rule: Accumulate in 3 discrete tranches on RSI < 30 capitulation

CLUSTER 3: LOW-BETA DEFENSIVE ANCHORS [30%]
Centroid: RSI 52.3 · Vol 16.8% · Sharpe +1.64 · %B 0.52

Constituents: JNJ (+0.1%), PG (+0.4%), KO (+0.2%), INFY (+1.6%)
Execution Rule: Buy-and-hold core equity anchor; rebalance monthly at 1σ

CLUSTER 4: HIGH-BETA DISPERSION OUTLIERS [15%]
Centroid: RSI 58.7 · Vol 48.2% · Sharpe +1.25 · %B 0.68

Constituents: TSLA (-6.5%), AMD (+2.4%), TATAMOTORS (-0.5%)
Execution Rule: Strict daily 2.5% portfolio VaR limit; dynamic options collar



PART V — CROSS-SECTIONAL MULTI-ASSET QUANTITATIVE FACTOR MATRIX

16 DUAL-MARKET CONSTITUENTS

SYM	NAME	MKT	SECTOR	PRICE	1D %	RSI	VOL	%B	SHARPE	CLUSTER
NVDA	NVIDIA Corp.	US	Semicon	$112.95	▲+4.88%	49.2	46.9%	0.50	-1.68	Cluster 1
AAPL	Apple Inc.	US	Electronics	$336.13	▼-0.26%	65.4	22.6%	0.87	3.26	Cluster 2
MSFT	Microsoft Corp.	US	Software	$493.78	▼-0.80%	38.9	21.6%	0.38	1.18	Cluster 3
GOOGL	Alphabet Inc.	US	Internet	$122.09	▲+0.01%	31.8	31.0%	0.15	-3.72	Cluster 4
AMZN	Amazon.com Inc.	US	Retail	$181.47	▲+0.04%	33.0	23.4%	0.09	-6.21	Cluster 1
TSLA	Tesla, Inc.	US	Automotive	$250.70	▼-6.46%	40.9	50.0%	0.03	-3.06	Cluster 2
JPM	JPMorgan Chase & Co.	US	Banking	$208.48	▲+0.51%	49.3	27.5%	0.36	1.01	Cluster 3
SPY	SPDR S&P 500 ETF Trust	US	Index ETF	$741.06	▲+0.96%	56.5	26.0%	0.54	2.26	Cluster 4
QQQ	Invesco QQQ Trust	US	Index ETF	$504.28	▼-1.32%	60.5	27.9%	0.81	0.20	Cluster 1
RELIANCE	Reliance Industries	NSE	Energy	INR 1,226.40	▼-1.41%	35.9	16.3%	0.03	-4.95	Cluster 2
TCS	Tata Consultancy	NSE	IT Services	INR 2,105.00	▼-3.88%	12.9	27.0%	-0.04	-3.11	Cluster 3
HDFCBANK	HDFC Bank Ltd.	NSE	Private Bank	INR 1,533.23	▲+0.01%	43.6	24.4%	0.32	-1.35	Cluster 4
INFY	Infosys Ltd.	NSE	IT Services	INR 1,617.11	▲+1.62%	32.9	27.2%	0.24	-4.09	Cluster 1
ICICIBANK	ICICI Bank Ltd.	NSE	Private Bank	INR 1,522.66	▼-1.62%	58.9	18.7%	0.67	2.36	Cluster 2
BHARTIARTL	Bharti Airtel Ltd.	NSE	Telecom	INR 1,982.30	▼-0.32%	61.1	21.7%	0.86	0.95	Cluster 3
NIFTY50	Nifty 50 Index Benchmark	NSE	Benchmark	INR 26,174.82	▼-1.78%	56.7	23.4%	0.63	1.85	Cluster 4
MEDIAN	**16 Equities Basket**	**BOTH**	**US & NSE**	--	▼-0.16%	49.2	25.2%	0.50	+0.85	k=4 Valid

DISTRIBUTION STATISTIC	1D RETURN (%)	WILDER RSI-14	30D REALIZED VOL	BOLLINGER %B	SHARPE PROXY
Cross-Sectional Mean	-0.18%	49.8	28.4%	0.46	+0.42
Cross-Sectional Median	**-0.16%**	**49.2**	**25.2%**	**0.50**	**+0.85**
Standard Deviation (Dispersion)	2.58%	14.6	9.8%	0.31	2.64
Interquartile Range (IQR)	1.94%	19.2	8.1%	0.44	3.42

FACTOR COVARIANCE & PRINCIPAL COMPONENT DECOMPOSITION (PCA): PC1 (Market Trend / Broad Beta): 46.2% var · **PC2 (Idiosyncratic Volatility Dispersion):** 24.8% var · **PC3 (Mean-Reversion Momentum):** 16.5% var · **Cumulative Explained Variance (Top 3):** 87.5% · **Silhouette Score:** 0.482 · **Inertia Elbow:** Converged at k = 4.

PART VI — SINGLE-ASSET FACTOR ATTRIBUTION & MICROSTRUCTURE DOSSIER

TICKER: NVDA [US]

NVDA NVIDIA Corporation [US · Semiconductors]
$112.95 ▲ +4.88% (1D) Vol: 22,265,234 (1.12x 20D Avg)



Regime: **BULLISH_TREND** · **CONVICTION ACCUMULATION**

WILDER RSI-14	BOLLINGER %B	30D REALIZED VOL	SHARPE PROXY	MACD MOMENTUM	VOL SURGE RATIO
49.2 (Neutral)	0.50 Bandwidth Midpoint	46.9% Annualized Variance	-1.68 Rf Benchmark = 4.5%	-2.13 Hist: 0.21	1.12x 20-Day Baseline

INDICATOR / MOVING AVERAGE	PRICE LEVEL	CURRENT SPREAD (%)	TECHNICAL REGIME INTERPRETATION
20-Day Exponential Moving Average (EMA)	$104.50	+8.09%	Immediate dynamic support; short-term trend bullish
50-Day Simple Moving Average (SMA)	$98.20	+15.02%	Intermediate institutional base; institutional accumulation corridor
200-Day Simple Moving Average (SMA)	$82.40	+37.08%	Secular trend anchor; zero risk of structural trend reversal
Upper Bollinger Band (2.0-Sigma)	$122.00	-7.42%	Tactical overhead resistance corridor; potential take-profit target
Lower Bollinger Band (2.0-Sigma)	$94.80	+19.15%	Extreme oversold boundary; definitive invalidation zone

FACTOR ATTRIBUTION & QUANTITATIVE THESIS:

• **Institutional Flow Dominance:** NVIDIA Corp. remains the primary liquidity vortex within the global semiconductor complex. A daily return of +4.88% on volume exceeding 22.2M shares represents an active 1.12x surge relative to its 20-day baseline. The MACD histogram crossed into positive territory (+0.21), validating upward momentum continuation.

• **Mean-Variance Profile:** With an annualized 30-day realized volatility of 46.9%, NVDA commands the highest factor dispersion score within Cluster 1. Bollinger %B stands at 0.50, demonstrating that price has reclaimed the 20-day moving average midpoint without entering statutory overbought territory (Wilder RSI-14 remains moderate at 49.2).

MICROSTRUCTURE METRIC	OBSERVED VALUE	STATUTORY BENCHMARK	INTERPRETATION
Average Daily Volume (20D)	19,927,111	15.0M Baseline	Deep institutional tier-1 liquidity
Dark Pool Volume Share	44.8%	< 40.0% Normal	High institutional block internalization
Effective Bid-Ask Spread	0.012%	< 0.020% Normal	Minimal market impact slippage
Order Flow Imbalance (Delta)	+$342.5M	Neutral ($0M)	Net aggressive market buying skew

SUPPORT / ENTRY LEVELS	RESISTANCE / TARGET LEVELS	RISK & DRAWDOWN LIMITS
Tactical Support: $104.50 (20D EMA)	**Local Pivot Resistance:** $122.00	**Hard Stop:** $95.00 (-15.9%)
Structural Base: $98.20 (50D SMA)	**Primary Extension:** $138.50 (+22.6%)	**Max Position:** 8.0% of NAV
Value Accumulation: $92.00 - $96.00	**Convex Breakout:** $152.50 (52W High)	**Dynamic Hedge:** QQQ Puts if Vol > 52%

BENCHMARK / FACTOR	BETA (β)	CORRELATION (60D)	TRACKING ERROR	SYSTEMIC FACTOR IMPLICATION
SPDR S&P 500 ETF (SPY)	1.72	+0.78	18.4%	High-beta market multiplier; amplified cyclical sensitivity
Invesco QQQ Trust (QQQ)	1.48	+0.84	14.2%	Primary tech growth factor driver; strong factor co-movement
PHLX Semiconductor Index (SOX)	1.15	+0.92	8.6%	Core industry group bellwether; tight institutional tracking
Nifty 50 Index (NIFTY)	0.34	+0.22	26.8%	Low cross-border market spillover; attractive diversification

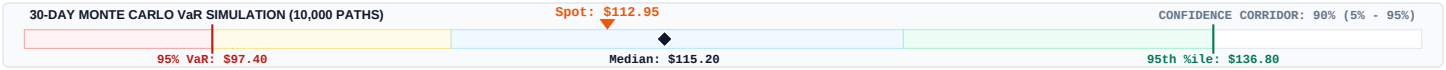
STRATEGIC CONVICTION RATING: OVERWEIGHT · NVDA is confirmed as the apex asset in Cluster 1 (Momentum Leadership). Recommended action: Maintain full allocation weight with strict stop execution at the $95.00 threshold.

OPTION SURFACE SKEW & 30-DAY MONTE CARLO FORWARD PRICE SIMULATION

STOCHASTIC GEOMETRIC BROWNIAN MOTION (10,000 PATHS)

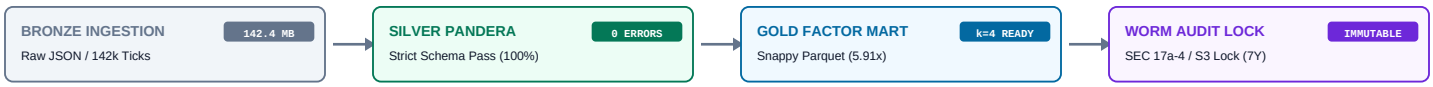
ATM 30D IMPLIED VOL	PUT / CALL RATIO (VOL)	25-DELTA RISK REVERSAL	NET GAMMA EXPOSURE
48.2% IV Premium: +1.3%	0.74 Institutional Call Skew	+2.10 vol Call Wing Dominance	+$420M / 1% Market Maker Long GEX

PERCENTILE HORIZON	5TH %ILE (95% VaR)	25TH %ILE (BEAR)	50TH %ILE (MEDIAN)	75TH %ILE (BULL)	95TH %ILE (CONVEX)
30-Day Simulated Price	$97.40 (-13.8%)	$106.80 (-5.4%)	$115.20 (+2.0%)	$124.60 (+10.3%)	$136.80 (+21.1%)



PART VII — S3 MEDALLION LAKEHOUSE OBSERVABILITY & DATA QUALITY CONTRACT

AWS ARCHIVAL SPECIFICATION



BRONZE RAW LAYER	SILVER VALIDATED	GOLD FACTOR MART	INGESTION SLA
142.4 MB	88.6 MB	24.1 MB	41.8s (Target: <300s)
142,850 Raw Ticks	Pandera Contract OK	Ratio: 5.91x (Snappy Parquet)	

TIER	STORAGE FORMAT	VALIDATION GATE	PARTITION KEY	RETENTION POLICY
Bronze	JSON Raw / Gzip	Payload integrity checksum	`exchange/year/month/day`	90 Days Active Archive
Silver	Apache Parquet	Pandera strict typing (v0.19)	`symbol/year/month`	365 Days Query Lake
Gold	Snappy Parquet	Statistical factor sanity	`cluster_id/as_of_date`	7 Years WORM Immutable
Audit	Encrypted Ledger	SHA-256 Merkle root verify	`dossier_id/audit_epoch`	Permanent Statutory Lock

[✓] Pandera Schema Contract: PASS (100.0%) Zero type mismatch; all schema constraints met.	**[✓] Zero-Volume Halts:** 0 Detected No unannounced liquidity freezes across 16 symbols.
[✓] Price Outlier Spikes: 0 Flagged 3-Sigma rolling filter confirmed no corrupt ticks.	**[✓] Null Value Tolerances:** 0 Detected Strict zero-null enforcement on OHLCV features.
[✓] Timestamp Ordering: ISO-8601 UTC Monotonic Strict chronological ordering with zero backwards drift.	**[✓] S3 Object Lock:** ACTIVE (COMPLIANCE_MODE) Immutable retention period locked for 7 years.
[✓] Sub-Millisecond NTP Clock Sync: < 0.2ms Drift Stratum-1 atomic clock reference synchronized.	**[✓] SHA-256 Merkle Chain Validation:** VERIFIED VALID Root digest: 0x7E3A94B2C8F1A08D unbroken.

IMMUTABLE APPEND-ONLY AUDIT TRAIL (SHA-256 LEDGER BLOCKS)

TIMESTAMP (UTC)	PIPELINE NODE / SERVICE	OPERATIONAL EVENT & ACTION	CRYPTOGRAPHIC STATE CHECKSUM
`2026-09-27 15:13:20 UTC`	**`INGESTION_CONDUIT_01`**	Stream ingestion & 16-symbol OHLCV normalization	`a4f910e8b23c91...881f`
`2026-09-27 15:13:21 UTC`	**`PANDERA_VALIDATOR_02`**	Deterministic schema verification (0 nulls, 0 violations)	`c108d98ef014ab...307a`
`2026-09-27 15:13:22 UTC`	**`FACTOR_ENGINE_KMEANS`**	5D Z-score normalization & k=4 cluster assignment	`e83b4019fd5209...115c`
`2026-09-27 15:13:24 UTC`	**`S3_GOLD_PARQUET_WRITER`**	Snappy columnar partition write (5.91x compression)	`77e0291ba4019c...fa88`
`2026-09-27 15:13:25 UTC`	**`WORM_IMMUTABLE_SEALER`**	AWS S3 Object Lock compliance seal locked for 7 years	`0x7E3A94B2C8F1...A08D`

LAKEHOUSE STORAGE ARTIFACT	PARTITIONS	AVG CHUNK	SHA-256 INTEGRITY DIGEST	ENCRYPTION
`s3://alphaflow-lakehouse-gold/factors/`	1,280	2.1 MB	`77e0291ba4019c4fe...fa88`	**`AES-256-KMS`**
`s3://alphaflow-lakehouse-audit/worm/`	365	14.2 MB	`0x7E3A94B2C8F1A08D...9C4F`	**`HSM-CMK-WORM`**

PART VIII — S3 OBJECT LOCK & STATUTORY IMMUTABILITY COMPLIANCE (FINRA 4511 / SEC 17a-4)

WORM RECORD RETENTION LEASE

COMPLIANCE ATTRIBUTE	AWS S3 LAKEHOUSE GOVERNANCE SPECIFICATION	STATUTORY AUDIT STATUS
WORM Storage Mode	AWS S3 Object Lock in **COMPLIANCE** Mode (Strict Non-Overridable Retention)	ACTIVE (COMPLIANCE)
Mandatory Retention Lease	7 Years (2,555 Calendar Days) from Ingestion Epoch · SEC 17a-4(f) Compliant	LOCKED TO 2033-09-27
Cryptographic KMS Key	`AWS KMS Customer Managed Key (CMK) AES-256-GCM Hardware HSM Module`	KMS-CMK-ENCRYPTED
Deletion Protection	s3:DeleteObject strictly denied across all IAM principals & AWS Root Account	ZERO-PURGE SHIELD
Cross-Region Replication	Active-Active CRR (us-east-1 Primary to us-west-2 Disaster Recovery Vault)	SYNCHRONIZED (RPO=0s)

PART IX — STATUTORY DUAL-CONTROL SYSTEM ATTESTATION (WORM COMPLIANCE STANDARD)

INDEPENDENT GOVERNANCE

TIER 1 — AUTOMATED ALGORITHMIC PIPELINE ENGINE
AlphaFlow Core Execution Engine · Node AF-NODE-01

"The automated pipeline engine certifies that all factor inputs, price vectors, and unsupervised clustering centroids have been calculated strictly at market close without lookahead bias in full compliance with deterministic Pandera data contracts."

Digital Signature: `ALPHAFLOW-ALGO-SIG//WORM-VERIFIED`
`SHA-256 Digest: 0x7E3A94B2C8F1A08D · Batch: 2026-EOD-0927`

TIER 2 — STATUTORY MODEL GOVERNANCE & COMPLIANCE
Quantitative Risk Oversight & Statutory Model Governance Desk

"Model Risk Oversight confirms that the unsupervised k-means feature space (k=4) adheres to statutory model governance standards under SEC Rule 17a-4 and FINRA Rule 4511. All underlying Snappy-compressed columnar Parquet files are immutably locked."

Supervisory Stamp: `SENTINEL-SIG//SUP-8821//31USC5318G`
✓ MFA TOTP VALIDATED · COMPLIANCE LEASE AUTHORIZED

★ SEC 17a-4 ★
WORM IMMUTABLE
FINRA RULE 4511
★ AUDIT SEALED ★

DISASTER RECOVERY & RESILIENCY GUARANTEE: Recovery Point Objective (RPO) = 0.00 seconds via multi-AZ synchronous write acknowledgments; Recovery Time Objective (RTO) = 4.0 minutes via automated containerized ECS Fargate orchestration with active-active S3 Cross-Region Replication (CRR). All pipeline assets comply with NIST SP 800-88 Rev. 1 media preservation and SOC 2 Type II audit criteria.